

# HEXBOARD SEQUENCER MODE QUICK MANUAL

## Overview

Sequencer mode is a separate mode from Keyboard mode. It is a single-track, 32-step sequencer. Each step can hold up to 4 notes, its own note length, its own velocity, its own probability, and can play through MIDI or the onboard synth.

## Enter And Exit

- From the main menu, choose Sequencer.
- In the Sequencer menu, choose Keyboard to return to normal keyboard mode.

## Main Controls

- Steps: the first 8 playable keys of the first 4 rows are steps 1-32.
- Play/Stop: the top-right key starts and stops playback on a short press.
- Play/Stop also deselects the current step if one is selected.
- Performance monitor: hold the top-right Play/Stop key for 2 seconds to show audio engine load, memory, storage, and MIDI receive stats; release it to close the monitor.
- Overview: the 9th key on the second row pages through an 8-step note view.
- Blue action key: the far-right key on the second row does undo on short press and clears the selected step on a 1-second hold.
- Tools: the last button on the third row opens a step-tools picker for the selected step.
- Encoder:
  - Turn while a step is selected to change that step's length quickly.
  - Press during file or folder naming to save the name.

## Step Editing

- Press a step once to select it.
- Press the same step again to deselect it.
- With a step selected, play note keys on the lower part of the board to add or remove notes from that step.
- The edit screen shows a compact info line for the selected step: ``L ...% V ... P ...%``, then the notes.
- Step notes are kept sorted from lowest to highest.
- Tap Preview can make a tapped step audition its stored notes when selected.

## Step Tools Picker

- Press the last button on the third row with a step selected to open tools.
- The tools screen shows the selected step number, a compact ``L ...% V ... P ...%`` line, and the current notes.
- Current tool choices are:  
Len, Vel, Oct+, Oct-, Prob, Tie
- The tools screen also shows:  
Cancel/Finished
- Len opens exact length entry for the selected step.
- Oct+ transposes the selected step up 1 octave and stays in the tools screen.
- Oct- transposes the selected step down 1 octave and stays in the tools screen.
- Vel opens exact velocity entry for the selected step.
- Prob opens exact probability entry for the selected step.
- Tie is still a placeholder entry for now.
- Cancel/Finished closes the picker when you are done.

## Step Velocity

- Velocity is stored per step.
- Range is 0-127.
- Default step velocity is 96.
- Use Vel from the step tools picker to edit it.
- Velocity entry uses the encoder.
- Turn the encoder to move through:  
0, 5, 10, 15, ... 125, 127
- Press the encoder to save.
- Playback, tap preview, save, and load all use the stored step velocity.

## Step Probability

- Probability is stored per step.
- Range is 0-100%.
- Default step probability is 100%.
- Use Prob from the step tools picker to edit it.
- Probability entry uses the encoder.
- Turn the encoder to move through:  
0, 5, 10, 15, ... 100
- Press the encoder to save.
- Playback, save, and load all use the stored step probability.
- Tap Preview ignores probability and still auditions the step.

## Note Length

- With a step selected, turn the encoder for quick length changes.
- Use Len from the step tools picker to open exact length entry.
- Available lengths are:  
0%, 25%, 50%, 75%, 100%, 150%, 200%, 250%, 300%, 350%, 400%, 500%, 600%, 700%,  
800%, 900%, 1000%

## Exact length screen:

- press number keys to enter the value
- the first digit typed replaces the current value
- up to 4 digits are allowed
- <- deletes the previous digit and is on the third-row first key
- CANCEL exits without saving and is on the fifth-row first key
- press the encoder to save

## Playback

- Tempo is the step clock.
- Each step is a 16th note at the current tempo.
- Hold Play/Stop for 2 seconds to show the performance monitor while held.
- Steps sets the active playback range from 1 to 32.
- Step lights beyond the active Steps count stay off, even if those steps still contain notes.
- Direction choices are:  
Forward, Backward, Ping-Pong, Random, Brownian, Drunk
- Play Type chooses:  
MIDI or OB Synth

## Menu Items

- Save: saves to the current file.
- Save New: saves as a new file in the selected folder.
- Load: loads a saved sequence.
- Revert: reloads the last saved version of the current sequence.
- File Management:  
Rename File, Rename Folder, Create Folder, Delete File, Delete Folder
- Steps: active playback length

- Direction: playback order
- Tempo: BPM
- Tap Preview: audition step contents when a step is tapped
- Play Type: MIDI or onboard synth
- Synth Options: opens the shared synth settings page
- Update Firmware: reboots to the firmware update bootloader

## Saving And Files

- Sequences are stored under /Sequences in onboard flash.
- On startup, the sequencer restores the last loaded sequence file when that remembered path is still valid.
- File browsers show up to 8 file/folder rows at a time.
- `^ more ^` and `v more v` indicate earlier or later pages.
- Scrolling past the first or last real file row pages automatically.
- The menu title shows the current file name as:  
Seq-FILENAME
- If there are unsaved changes, it becomes:  
Seq-\*FILENAME
- If no file is loaded, the title just says:  
Sequencer

## Naming Mode

When naming a file or folder, the board shows this entry layout:

```
A B C D E F G H
I J K L M N O P
Q R S T U V W X
Y Z SPC - 1 2 3
<- CANCEL
```

- Press keys to enter characters.
- SPC adds a space.
- <- deletes the previous character.
- CANCEL exits naming without saving.
- Press the encoder to save the name.
- the physical CANCEL key is placed lower on the board to reduce accidental exits

## Status And Display

- The selected step blinks.
- The overview screen shows 8 steps at a time.
- Empty steps are dark when idle.
- The performance monitor shows:  
AudioEng, Mem, FS, MIDI Q, Drop, and Late while the Play/Stop key is held.
- Active sequencer overlays stay awake while you are using them.
- The top title shows the loaded file name and unsaved state.